

Outage Investigation Report —

OIR-2025-004

Field	Value
Document ID	OIR-2025-004
Report Date	2025-12-27
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Reviewing Manager	Priya Ramachandran, Plant Manager, NW-PowerPool
Region	NW-PowerPool
Site	NW-PowerPool-WIND-003
Asset	TRA-10561 (Siemens Energy, transformer)
Outage ID	OUT-2025-1407
Classification	Forced outage — overheating
Distribution	NW-PowerPool Operations, Reliability Engineering, OEM Liaison, Compliance

1. Outage Summary

TRA-10561, the main step-up transformer feeding the NW-PowerPool-WIND-003 collector, tripped on top-oil over-temperature on 2025-12-10 at 10:38 PST. Restoration completed 2025-12-19 at 09:38 PST, total duration **215 hours**. Lost generation: **3,558.6 MWh**.

This is the second multi-day outage on TRA-10561 in 2025; the first was a vibration_excess event on 2025-05-26 (OIR-2025-007, 158 h, 2,615.1 MWh). The asset has now logged 8 forced outages and 7,900 MWh of lost generation in FY2025. Cumulative pattern is the subject of RCA-2025-002 (Fall Overheating Sequence) and RCA-2025-005 (procedural shortcut).

2. Timeline of Events

Timestamp (PST)	Event
2025-12-09 18:10	Top-oil temperature trending +0.6 °C/hr under 32 MVA load and ambient -2 °C.
2025-12-10 02:40	Top-oil crosses 75 °C "monitor" threshold; cooling fan banks 2 and 3 commanded on. Bank 3 fails to start (later confirmed: locked rotor on fan 3-D).
2025-12-10 08:55	Top-oil at 92 °C. Operator (M. Sandoval) requests load shed to 24 MVA. Pickup is delayed 12 minutes by NW-PowerPool dispatch.
2025-12-10 10:38	Top-oil over-temperature trip at 99.4 °C. Forced outage declared.
2025-12-10 12:00	CPS unable to mobilize within 4 hours (concurrent NW-PowerPool-WIND-024 work on WIN-10011 — see OIR-2025-009). EPA-2024-11 invoked.
2025-12-10 16:20	Apex onsite. Cooling fan bank 3 isolated. Dissolved gas analysis ordered.
2025-12-11 — 2025-12-17	DGA results elevated — acetylene 14 ppm (limit 1), ethylene 84 ppm, methane 41 ppm. Apex recommends transformer replacement. Internal engineering counter-proposes hot-oil cleaning, gasket replacement, and fan bank rebuild.
2025-12-18 — 2025-12-19	Field repair executed per internal engineering plan. Re-energization successful at de-rated 28 MVA pending follow-up DGA in 30 days.

3. Telemetry Observations

- Top-oil at trip: 99.4 °C (limit 95 per Siemens Energy Manual SET-2200 §4.3; Cascadia internal de-rating policy at NW ambient < 0 °C lowers to 90 °C — pending acceptance, see CAR-2025-005).
- DGA at fault: acetylene 14 ppm, ethylene 84 ppm, methane 41 ppm. Pattern is consistent with localized thermal fault, not arcing.
- Fan bank 3 lock-rotor: fan 3-D motor windings shorted; fans 3-A, 3-B, 3-C operative. The June 2025 PM (WO-2025-06035) recorded "all fans operational" without resistance check on individual motors.
- Ambient: -2 to -7 °C, sustained NW wind 7 m/s.
- Vibration baseline on the transformer tank shows a 12% increase since May 2025 (OIR-2025-007), indicating cumulative core looseness.

4. Restoration Activities

Vendor	Scope	Invoice (USD)
Apex Emergency Grid Solutions	Onsite diagnostics, DGA management, fan bank isolation	\$148,000
Apex / Siemens Energy subcontract (1.8x)	Hot-oil cleaning, gasket replacement, fan bank 3 rebuild	\$264,000
Total		\$412,000

CPS scheduled equivalent estimate (post-event benchmark): \$108,400. The 4x factor recurs in EMR-2025-005 and is summarized in RCA-2025-001.

5. Preliminary Root Cause

Direct cause: cooling fan bank 3 failure under cold-ambient peak-load conditions caused top-oil temperature excursion and DGA-indicated thermal fault.

Contributing causes:

1. Fan-motor resistance check omitted from the June 2025 PM (WO-2025-06035). The CSM-7-equivalent procedure for transformer auxiliaries was not executed; the technician's log notes "deferred to year-end shutdown" — see RCA-2025-005 and CAR-2025-006.
2. Top-oil de-rating policy for sustained cold ambient is documented in Standard MS-2025-03 but not loaded into the SCADA alarm template for TRA-10561. The unit was running to the OEM threshold rather than the internal threshold. See CAR-2025-005.
3. Cumulative core vibration since May (OIR-2025-007) was not actioned in the intervening 6 months — the planned mid-year inspection (WO-2025-08019) was deferred.

6. Recommendations

1. Execute fan-motor resistance checks on all NW-PowerPool transformer auxiliaries within 30 days. Track in CAR-2025-008.
2. Load the MS-2025-03 cold-ambient de-rating curve into SCADA alarm templates fleet-wide. Track in CAR-2025-005.
3. Reject Apex's transformer replacement recommendation. Internal engineering plan succeeded; replacement quote at \$1.86M not warranted.

4. Add TRA-10561 to FY26 capital planning watchlist; cumulative pattern indicates end-of-economic-life within 2 years.
5. Cross-reference OIR-2025-007 (May vibration event), RCA-2025-002 (Fall Overheating Sequence), RCA-2025-005 (procedural shortcut).

Signatures

Role	Name	Date
Author	Elena Vasquez, Reliability Engineer II	2025-12-27
Plant Manager	Priya Ramachandran	2025-12-28
Compliance Review	D. Whitcombe	2025-12-29

Distribution: NW-PowerPool Operations · Reliability Engineering · OEM Liaison · Compliance · Vendor Management · CFO Office

References: OIR-2025-007 · OIR-2025-009 · CAR-2025-005 · CAR-2025-006 · CAR-2025-008 · EMR-2025-005 · RCA-2025-002 · RCA-2025-005 · Siemens Energy Manual SET-2200 · Standard MS-2025-03 · CSM-7